

Jonathan Alves

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SUMMARY

Transitioning into a Developer role after an 8-year career in civil engineering and project planning. Building a strong Computer Science foundation at 42 Lisboa, focusing on Data Structures and Algorithms. Offering analytical precision, Fintech domain knowledge, and Fluent English. Proven ability to design, implement, and write clean, scalable code, perform rigorous debugging, and apply problem-solving skills to software products within cross-functional teams. Eager to work in a Junior/Intern developer position.

EDUCATION

42 Lisboa

Lisbon, Portugal

Software Engineering Student

01/2026 – Present

- Mastering software engineering fundamentals, Data Structures and Algorithms, and low-level programming in C and Python.
- Implementing strict design patterns (OOP), memory management, and scalable software architectures.
- Setting up repository automations with GitLab CI/CD for testing protocols, Makefiles, and strict code linting rules in the projects.
- Successfully completed the intensive 26-day 42 Piscine coding bootcamp.

FIAP - Paulista Faculty of Informatics and Administration

Sao Paulo, Brazil

Bachelor's in Information Technology Management

01/2017 – 03/2019

- Bridged technical development platforms with IT project methodologies (Agile, Scrum, Cobit).
- Awarded 3rd place for the best Artificial Intelligence project in the first year and 4th place finalist for the final thesis.

PROFESSIONAL EXPERIENCE

Business Analyst and Co-founder

06/2021 – 01/2025

Empvion Innovation Consulting

Lisbon, Portugal

- Focus on improving operational efficiency through strong data analysis and digital market trends.
- Map requirements, design new software systems and IT Solutions, optimizing internal process workflows for tech-oriented business models.

Senior Risk and Fraud Analyst

09/2022 – 09/2024

Cognizant (Fintech Domain)

Lisbon, Portugal

- Analyzing existing programs and investigating complex incidents using advanced data analysis tools to detect and mitigate suspicious patterns in digital wallet transactions.
- Collaborate directly with cross-functional multidisciplinary engineering and compliance teams to develop systemic solutions and maintain security systems.

Customer Support Representative

05/2018 – 08/2019

Uber

Sao Paulo, Brazil

- Performed deep troubleshooting of critical problems, evaluating user feedback, and fixing bugs reported on the platform.
- Created data-driven internal projects and technical documents to improve operational KPIs in collaboration with the Center of Excellence.

Cost Estimator and Technical Planner

01/2008 – 01/2017

Civil Engineering Sector

Sao Paulo - Rio, Brazil

- Managed and manipulated robust databases via ERP System and engineering software, ensuring extreme precision and version control.
- Applied analytical structuring and mathematical modeling to create, analyze and coordinate budgets, demonstrating the ability to work well under pressure.

PROJECTS

- Libft (Custom C Library)** | *C, Makefile, Unit Testing* 2026
- Designed and implemented a custom C standard library from scratch, demonstrating advanced memory management, pointer manipulation, and rigorous debugging practices.
 - Applied industry best practices for version control using Conventional Commits (cz-git) and structured Makefiles for efficient compilation.
- get_next_line (File Parsing and Memory Management)** | *C, GitLab CI/CD, Valgrind* 2026
- Engineered a file-descriptor line-reading function, achieving a **100/100 evaluation score** with **zero memory leaks** detected via Valgrind.
 - Built a GitLab CI/CD pipeline with **4 automated stages** (Norminette check, parallel BUFFER-SIZE builds, output validation, Valgrind memory check) running on every commit.
 - Authored comprehensive technical documentation (README) utilizing shields.io badges to communicate project architecture.
- push_swap (Sorting Algorithm)** | *C, Algorithms, Data Structures* 2026
- Implemented an tightened sorting algorithm with two stacks and a restricted instruction set, then validated it against 42's standard evaluation benchmarks on 100 / 500 / 5000-element datasets.
 - Engineered robust input parsing (**ft_atoi**) with rigorous range validation to prevent integer overflow vulnerabilities.
- GitFit (Official 42 Lisboa Club)** | *Founder and Organizer* 07/2026 – Present
- Founded and lead the official athletic group at 42 Lisboa, organizing monthly fitness challenges and a tiered badge ranking system (Bronze/Silver/Gold/Platinum) to promote well-being among developers.

SKILLS

Languages: Python, C, SQL, Bash

Libraries: Pandas, NumPy, Matplotlib, Pydantic

Tools: Git, GitHub, GitLab CI/CD, Conventional Commits, Valgrind, Docker, Linux, Jira, Trello, ClickUp, Figma

Architecture and Concepts: Object-Oriented Programming (OOP), Data Structures and Algorithms, API

Integration, Unit Testing, Clean Code, Agile development/Scrum, BPMN

Soft Skills: Analytical Thinking, Problem-solving, Fast Learner, Effective Communication, Resilience, Conflict Management

Languages Spoken: Portuguese (Native), English (Fluent – Professional Working Proficiency), Spanish (Basic), French (Basic)